

toyasm

Assembler and Simulator for the Toy CPU

A Python assembler, simulator, and GUI debugger
for Jim Hall's minimal 8-bit educational processor

Manfred Scheucher

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1 Concepts: how the Toy CPU works

If you have never written machine language before, this chapter is the foundation. Everything else builds on it.

1.1 What is a CPU?

A CPU (central processing unit) is a machine that does one thing over and over, extremely fast: it reads a number from memory, treats that number as an *instruction*, and does what the instruction says. Then it reads the next number, and the next, forever — until an instruction tells it to stop.

That is the whole idea. There is no magic. A program is just a list of numbers sitting in memory. The CPU walks through them one at a time. The Toy CPU is a deliberately tiny version of this idea, small enough to hold in your head all at once.

1.2 The three things you need to know about

The Toy CPU has exactly three pieces of state. That’s it.

1.2.1 Memory — 256 bytes

Memory is a row of **256 numbered boxes**, addressed 0 to 255. Each box (“byte”) holds one number from 0 to 255.

```
address:  0    1    2    3    4    5    ...  255
value:    [ 20 ] [ 7  ] [ 22 ] [ 8  ] [ 0  ] [ 5  ] ... [ 0  ]
```

Crucially, **memory holds both your program and your data**. There is no separate “code” and “data” — it’s all just numbers in boxes. The CPU can’t tell them apart on its own; *you* decide the layout, and the CPU trusts you.

1.2.2 The accumulator (ACC)

The accumulator is a single box *inside* the CPU where arithmetic happens. It holds 8 bits (a value 0–255). Almost every operation flows through it: to add two numbers you **load** one into the accumulator, then **add** the other; to save a result you **store** the accumulator back into a memory box. Think of the accumulator as your hands — you can only hold one value at a time, and you have to put it down (store it) before picking up another.

1.2.3 The program counter (PC)

The program counter is the CPU’s “you are here” marker. It holds the address of the *next* instruction to run. It starts at 0, so every Toy CPU program begins at memory address 0. After running an instruction the PC moves forward; **goto** and **ifzero** change it on purpose — that’s how loops and branches work.

1.3 The cycle: fetch, decode, execute

Everything the CPU does is this loop, repeated:

1. **Fetch** — read the byte at the address in the PC.
2. **Decode** — figure out which instruction that byte means (its *opcode*).
3. **Execute** — do it. If the instruction needs an address (like **add**), the CPU also reads the *next* byte to know which memory box to use.
4. **Advance** — move the PC forward, and repeat.

This continues until the CPU fetches a **stop** instruction.

1.4 One-byte and two-byte instructions

Some instructions are self-contained and take **one byte** (stop, right, left, not, nop). Others need to know *which memory box* to work with, so they take **two bytes**: the opcode, followed by the address.

```
load a    -> two bytes: [ 20 ][ 7 ]    "load, from address 7"
right     -> one byte:  [ 1 ]          "shift the accumulator right"
```

The CPU knows which is which from the opcode itself (a specific bit marks two-byte instructions — see the instruction set chapter).

1.5 What's missing (on purpose)

Compared to a real CPU, the Toy CPU leaves out almost everything: no multiply/divide, no multiple registers, no stack, no interrupts, no overflow flags. Values simply wrap around modulo 256. This is a feature — with so few parts you can understand *the entire machine*, and build up multiplication, comparison, and loops yourself from the primitives.

1.6 The ladder: from C down to the lights

On the original Toy CPU you enter a program the hard way: one byte at a time, in binary, by flipping switches and reading LEDs. That *is* the lesson — with no tools, **you are the assembler**, translating in your head. It teaches you what the machine really does, but it is slow and error-prone for anything longer than a few instructions.

toyasm adds layers on top that make programs easier to write, without ever hiding what the machine executes:

C source	(toycc)	int a = 7; return a * b;	most human
compile			
v			
assembly	(.toys)	load a / add b / stop	readable, named
assemble	(1:1)		
v			
machine code	(bytes)	[20][7][22][8][0]	what the CPU runs
run			
v			
Toy CPU		LEDs + switches	the bare machine

Two things make this honest rather than a black box:

- **Assembly maps 1:1 to machine code.** Each mnemonic is exactly one opcode byte; a label is just a name for an address. Assembling is a direct substitution — when you step through a program you can see each source line become the very bytes in memory.
- **The C layer is optional.** toycc compiles a small subset of C down to assembly (turning `a * b` into repeated addition, since the CPU has no multiply). It sits on top of the machine, not inside it — you can understand the Toy CPU completely without ever writing a line of C.

2 Instruction set

The Toy CPU understands 14 instructions. Each has a numeric **opcode** (the byte that identifies it). Instructions that reference memory are two bytes long (opcode + address); the rest are one byte.

Mnemonic	Dec	Binary	Bytes	Effect
stop	0	00000000	1	Halt execution
right	1	00000001	1	ACC = ACC >> 1 (shift right)
left	2	00000010	1	ACC = ACC << 1 (shift left)
not	15	00001111	1	ACC = ~ACC (bitwise NOT)
and	17	00010001	2	ACC = ACC & mem[addr]
or	18	00010010	2	ACC = ACC mem[addr]
xor	19	00010011	2	ACC = ACC ^ mem[addr]
load	20	00010100	2	ACC = mem[addr]
store	21	00010101	2	mem[addr] = ACC
add	22	00010110	2	ACC = ACC + mem[addr]
sub	23	00010111	2	ACC = ACC - mem[addr]
goto	24	00011000	2	PC = addr (jump)
ifzero	25	00011001	2	if ACC == 0: PC = addr
nop	128	10000000	1	Do nothing

Any opcode not in this table is treated as **nop**.

2.1 How the CPU knows the length

Look at **bit 4** (value 0x10, the 1 in 000 1 0000). If it is set, the instruction reads a second byte as its address. This is the *fetch bit*:

- **load** = 00010100 → bit 4 set → two bytes.
- **right** = 00000001 → bit 4 clear → one byte.

You never set this bit yourself — it’s baked into the opcodes. It is just how the CPU decides whether to read an address after the opcode.

2.2 Notes on individual instructions

- **Arithmetic wraps modulo 256.** $255 + 1 = 0$, $0 - 1 = 255$. There is no overflow flag. If a result must fit, keep values small.
- **right / left** shift the bits by one position, filling with 0. **left** is a quick multiply-by-2; **right** a divide-by-2 (dropping the low bit). Bits shifted off the end are lost.
- **ifzero** is the only conditional. Every “if” and every loop exit is ultimately built from **ifzero**.
- **goto** is an unconditional jump — used to loop back or skip ahead.
- **store** is how results leave the accumulator and become data in memory.

2.3 Building bigger operations

The Toy CPU has no multiply, divide, or compare instruction. You build them:

- **Multiply** → repeated **add**, or doubling with **left**.
- **Compare for equality** → **xor** two values; the result is 0 exactly when they’re equal, so follow it with **ifzero**.
- **Loop N times** → keep a counter in memory, **sub** one each pass, and **ifzero** to exit.

3 Assembly syntax

Programming the original Toy CPU meant flipping switches to enter raw binary (see the Toy CPU chapter). `toyasm` lets you write the same programs as readable text in a `.toys` file, and it does the tedious part — turning mnemonics and labels into the right bytes at the right addresses.

3.1 The shape of a program

```
# comments start with # and run to end of line
label: load var      # an instruction, optionally with a label
      add  other
      goto label

# data usually goes at the end
var:   42             # decimal
flags: 0b00001111    # binary (0b prefix)
mask:  0xFF           # hexadecimal (0x prefix)
bits:  00001111       # bare 8-bit binary is also accepted
```

3.2 Elements

Instructions are a mnemonic from the instruction set, optionally followed by an operand. Two-byte instructions (`load`, `add`, `goto`, ...) take one operand — a label naming a memory box. One-byte instructions (`stop`, `right`, `not`, ...) take none.

Labels name a memory address. Write `name:` before an instruction or a data value, then use the name anywhere an address is needed:

```
loop:  sub  one        # "loop" names this instruction's address
      ifzero end      # "end" and "one" are labels defined elsewhere
      goto loop
```

A label is just a human-friendly name for a number. `toyasm` figures out the actual address and substitutes it, so you never count bytes by hand.

Data is a label followed by a value; it reserves one byte holding that value. Because code and data share memory, put your data *after* the `stop` (or after a `goto`) so the CPU never tries to execute it as an instruction.

3.3 Number formats

Form	Example	Meaning
Decimal	42	Base 10
Hexadecimal	0xFF	0x prefix
Binary	0b00001111	0b prefix
Bare 8-bit	00001111	Exactly 8 0/1 digits, read as binary

All values must fit in one byte (0–255).

3.4 How `toyasm` lays it out in memory

Lines are placed into memory top to bottom, starting at address 0: a one-byte instruction takes one address; a two-byte instruction takes two (opcode, then the resolved address); a data line takes one address holding its value. So this source:

```
    load  a    # addr 0-1
    add   b    # addr 2-3
    stop                # addr 4
a:      10      # addr 5
b:      20      # addr 6
```

compiles to the bytes 20, 5, 22, 6, 0, 10, 20 at addresses 0–6. The label **a** resolves to 5 and **b** to 6, which is why `load a` becomes 20, 5.

You can see this layout for any program with the CLI's export option (`--cli --export`) or in the GUI's memory panel.

4 Writing programs

This chapter builds up from a straight-line program to loops, conditionals, and a couple of advanced tricks. Each example is in the `examples/` folder — run them and watch them step. Make sure you’re comfortable with the concepts first: memory boxes, the accumulator, the program counter, and that every calculation flows through the accumulator.

4.1 Straight-line: $a + b + c$

The simplest programs just do one thing after another. Load a value, combine in more values, stop. `sevenfold.toys` computes $7 * a$ with only adds:

```
load  a      # acc = a
add   a      # acc = 2a
store b      # b = 2a
load  b
add   b      # acc = 4a
store c      # c = 4a
load  a
add   b
add   c      # acc = a + 2a + 4a = 7a
stop

a:    4
b:    0
c:    0
```

The pattern to internalize: **load to start a calculation, add/sub to build it up, store to save an intermediate, stop when done.** The data (a, b, c) sits after `stop` so it’s never executed.

4.2 Loops and counters: multiply by repeated addition

The Toy CPU has no multiply instruction, so you build it. Keep a counter in memory, do work each pass, decrement the counter, and jump back until it hits zero. `multiply.toys` computes $a * b$:

```
loop:  load  a      # acc = a (used as loop counter)
       ifzero done # counter == 0 -> exit the loop
       load  sum
       add   b      # sum += b
       store sum
       load  a
       sub   one    # a--
       store a
       goto  loop   # back to the top

done:  load  sum      # result in accumulator
       stop

a:     7
b:     6
sum:   0
one:   1
```

The loop skeleton is worth memorizing — nearly every Toy CPU loop looks like it: `ifzero` at the top checks the exit condition; the body does the work; a counter is decremented with `sub one`; `goto` jumps back to the top.

4.3 Conditionals from `ifzero` + `xor`

`ifzero` is the *only* branch instruction, so every decision is phrased as “is this value zero?”. Two useful reductions:

- **Equality:** `xor` two values. The result is 0 exactly when they’re equal. Follow with `ifzero` to branch on “equal”.
- **Countdown to a condition:** `sub` until you reach 0, then `ifzero`.

`fibonacci.toys` uses the countdown form to run its loop `n` times:

```
      load  n
loop:  ifzero end      # done after n iterations
      ...             # (compute next Fibonacci number)
      load  n
      sub   one
      store n          # n--
      goto  loop
end:   load  a
      stop
```

4.4 Advanced: self-modifying code for arrays

The Toy CPU has no index register, so to walk through an array you modify the program itself while it runs — because code and data are the same memory, you can **store** a new address into an instruction’s operand byte.

`sum.toys` and `max.toys` do this. The key trick is writing a `load` as two raw bytes so the address byte is a normal, patchable memory box:

```
iop:   20              # LOAD opcode, as a raw data byte
iarg:  0               # <- the address byte, overwritten each pass
      add   result
```

Each iteration, the program does `load iarg; add one; store iarg` to bump `iarg` to the next array element. It’s a genuine machine-language technique — and a good demonstration of *why* code and data sharing memory is powerful and dangerous.

4.5 Tips for writing your own

- **Sketch on paper first.** Decide your variables and the loop shape before writing instructions.
- **Put all data after a `stop` or `goto`** so it’s never executed.
- **Watch for wraparound.** Values are mod 256; `fib(14) = 377` overflows, so `fibonacci.toys` only works up to `n = 13`.
- **Step through in the GUI** when something’s wrong — watching the PC move and the accumulator change makes bugs obvious.

5 Using toyasm

toyasm assembles a .toys file and then either opens a graphical interface (the default) or runs it in the terminal.

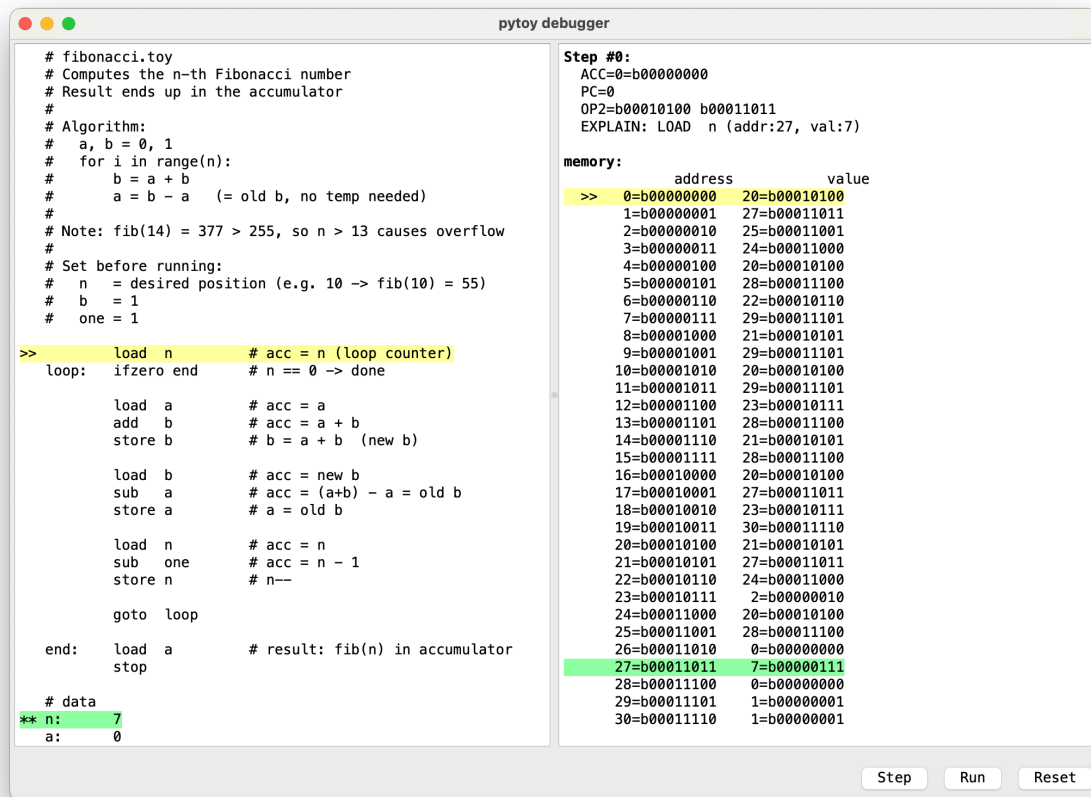


Figure 1: The toyasm GUI: source panel on the left (current instruction highlighted), memory panel on the right showing every byte in binary.

5.1 Requirements

- Python 3
- PySide6 — only for the GUI. The CLI works without it.

```
pip install PySide6
```

5.2 GUI (default)

Just point toyasm at a program:

```
python3 toyasm.py examples/fibonacci.toy
```

The window has:

- **Source panel** — your program, with the current instruction highlighted in yellow and any memory it references in green.
- **Memory panel** — every address and its value in binary, so you watch the bits change as the program runs.
- **Controls** — Step, Run, and Reset buttons.

Key	Action
Space	Step one instruction
R	Run to completion
Escape	Reset to the start

Click any line to highlight it (orange) on both panels — handy for tracing which memory box a label refers to.

5.3 CLI

Use `-c / --cli` to run in the terminal instead of opening the window:

```
python3 toyasm.py examples/fibonacci.toys --cli
```

By default the CLI runs step-by-step with full verbose output; press Enter to advance each instruction.

Flag	Effect
<code>-c, --cli</code>	Run in the terminal instead of the GUI
<code>-r, --run</code>	Run all steps without pausing
<code>-q, --quiet</code>	Compact, one line per step
<code>-x, --export</code>	Also write the compiled listing to a <code>.toyo</code> file

Run `python3 toyasm.py -h` for the full list.

```
# Run to the end, compact output:
python3 toyasm.py examples/multiply.toys --cli --run --quiet

# Assemble and export the byte layout to examples/max.toyo:
python3 toyasm.py examples/max.toys --cli --export --run
```

The final line reports the result in the accumulator in binary, hex, and decimal.

5.4 Memory limits and code-overwrite detection

The Toy CPU has exactly 256 bytes, shared by code and data. Two things can go wrong at that boundary, and `toyasm` handles both:

Too big to fit. If a program assembles to more than 256 bytes, the assembler stops with a clear message rather than crashing:

```
ERROR: Program is too big: it needs 402 bytes, but the Toy CPU has only 256.
```

The C compiler `toycc` performs the same check on its generated output, so an oversized C program is caught at compile time instead of failing later.

Overwriting code. Because code and data share memory, a `store` can write over your own instructions. That is a legitimate technique (the array examples use it deliberately — see the writing-programs chapter), so it is allowed by default. But when you *don't* mean to do it, it is a nasty bug. The optional `-d / --detect-code-overwrite` flag turns on a guard:

```
python3 toyasm.py myprog.toys --cli -d
```

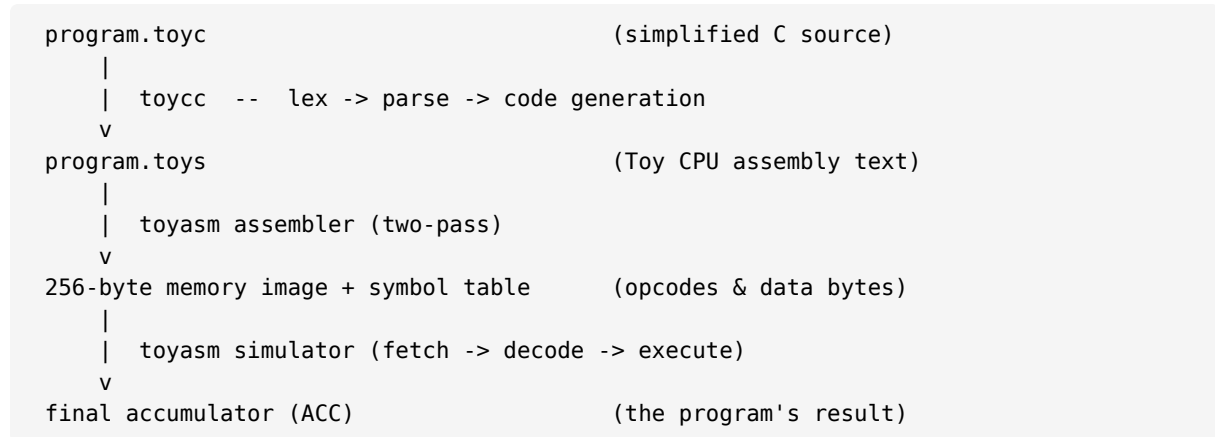
The guard uses the `# data` marker in your source — a line whose only content is `# data` — to know where the code region ends and data begins. If a `store` writes below that line (into code), `toyasm` warns and asks whether to continue; in the GUI it pops up a dialog. If the program has no `# data` marker, the guard can't know the boundary and prints a note that detection is off. The flag is off by default, keeping the bare machine's “anything goes” behaviour unless you opt in.

6 The C → assembly → CPU pipeline

toyasm ships with **toycc**, a small compiler that turns a simplified subset of C into Toy CPU assembly (**.toys**). Writing C is more pleasant than hand-writing assembly, and it lets you see, stage by stage, how a high-level program becomes raw machine bytes.

The compiler lives in `compiler/`; its README documents the exact supported C subset.

6.1 The stages



Stage	Tool	Input	Output
1. Compile	toycc	.toycc source text	.toys assembly text
2. Assemble	toyasm	.toys text	memory image + symbols, or errors
3. Simulate	toyasm	memory image	final ACC value

6.1.1 Stage 1 — toycc (C → assembly)

toycc is a hand-written compiler (standard library only): a **lexer** splits the source into tokens, a **recursive-descent parser** builds a syntax tree, and a **code generator** walks that tree emitting Toy CPU instructions. Because the machine is so small, code generation makes concrete choices you can inspect in the output:

- Every C variable gets a fixed **data byte** in memory; temporaries get their own bytes too.
- `a * b` becomes a **repeated-addition loop** (there is no multiply opcode).
- Comparisons reduce to the one available conditional, `ifzero` — e.g. `a == b` compiles to `a - b` followed by `ifzero`, and ordering uses the sign-bit-of-difference trick.
- `while` / `for` / `if` become `goto` and `ifzero` with generated labels.

6.1.2 Stage 2 — the assembler (assembly → bytes)

toyasm's two-pass assembler resolves labels to addresses and emits the 256-byte memory image. This is the same assembler used for hand-written **.toys** programs; toycc's output is nothing special to it. Invalid input yields errors instead of a memory image.

6.1.3 Stage 3 — the simulator (bytes → result)

The simulator runs the fetch–decode–execute cycle until `stop`, leaving the program's `return` value in the accumulator, printed as the final `Result: ACC = ...` line.

6.2 Running the pipeline by hand

```
# Stage 1: compile C to assembly
python3 compiler/toycc.py compiler/examples/multiply.toyc
```

```
# Stages 2+3: assemble and simulate
python3 toyasm.py compiler/examples/multiply.toys --cli --run --quiet
#   -> Result: ACC = 42

# Or all three at once with toycc's --run flag:
python3 compiler/toycc.py compiler/examples/multiply.toyc --run
```

6.3 How the pipeline is tested

The whole pipeline is exercised by automated tests in `tests/test_toycc.py`, run with `pytest`. The tests are **data-driven**: every `compiler/examples/*.toyc` file carries a machine-readable `// expect: N` annotation stating its correct result. For each example the harness runs all three stages (`compile_source #sym.arrow assemble #sym.arrow simulate`) and asserts the accumulator equals `N`. Adding a new example automatically adds a test.

Focused unit tests cover individual compiler features (arithmetic, bitwise ops, shifts, `if/else`, `while`, `for`, comparisons, wraparound) so a failure points at the specific feature. A git pre-commit hook (installed via `scripts/install-hooks.sh`) runs this suite before each commit.

7 About the Toy CPU

toyasm is a Python assembler and simulator for the **Toy CPU**, a minimal educational processor created by Jim Hall for the FreeDOS Project. This chapter records where the machine comes from and how toyasm relates to it.

- Official page: <https://jimhall.itch.io/toy-cpu> (free, MIT-licensed)
- Source: <https://github.com/freedosproject/toycpu>
- Jim Hall’s talk (VCF East, 50 min): <https://www.youtube.com/watch?v=zhoL1ZSjGfM>

7.1 The “switches and lights” heritage

Early microcomputers like the **Altair 8800** and **IMSAI 8080** (both 1975) had no keyboard or screen. You entered programs by flipping a bank of **switches** to set each binary instruction, and read results from a row of **LEDs** (“lights”). Programming meant knowing the raw opcodes by heart and toggling them in bit by bit.

The Toy CPU faithfully recreates this: a counter, an instruction, and an accumulator, each shown as 8 lights, with a tiny instruction set instead of the real Altair opcodes.

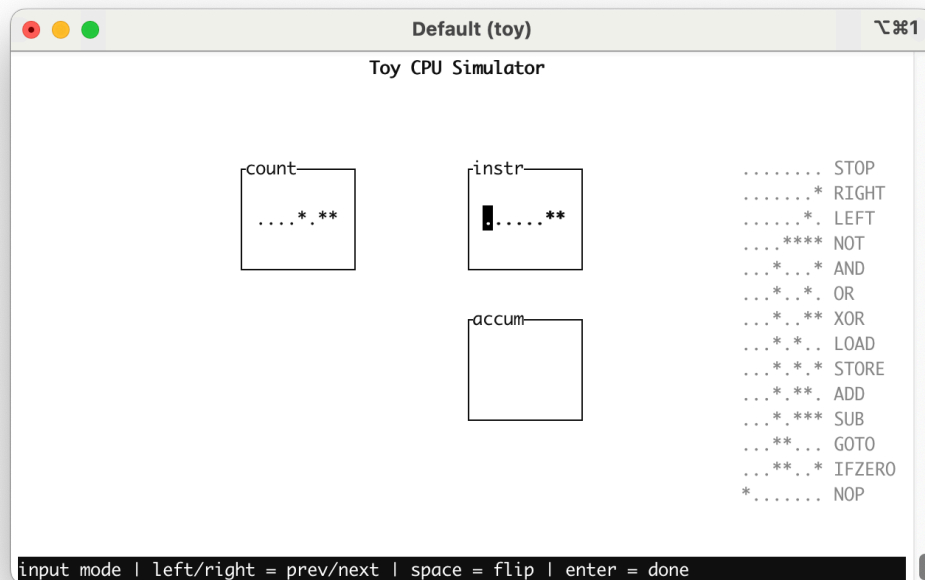


Figure 2: Jim Hall’s original Toy CPU (v2, ncurses). The `count`, `instr`, and `accum` boxes are 8-bit LED patterns; the legend on the right lists every opcode. You program it by flipping bits: *left/right* = *prev/next*, *space* = *flip*, *enter* = *done*. `toyasm` exists to spare you exactly this bit-by-bit entry, without hiding what it produces.

7.2 Why toyasm exists

The Toy CPU itself is wonderful because it is *small* — the whole machine fits in your head. But driving it through LEDs and switches gets tiring fast: entering a program bit by bit is slow, you can’t see much of what’s going on, and you tend to give up experimenting before you’ve really explored. That switch-and-light workflow has its own charm — it’s honest about how bare a computer really is — but it’s a narrow doorway.

The sharpest difference is what you can *see at once*. On the original front panel you look at one value at a time — one register, one address — and step or click through the rest; the machine only

ever shows you a single 8-bit box. `toyasm` shows everything simultaneously: the whole program, every byte of memory, the accumulator, and the program counter, all on screen together, updating as you step. Watching the PC move and the bytes change in one view is what makes a bug obvious instead of invisible.

So `toyasm` keeps the tiny, comprehensible machine and widens the doorway. With a graphical view of all of memory and the accumulator, an assembly layer, and a C layer on top, you spot your mistakes almost immediately (instead of staring at LEDs that tell you nothing) and you can keep experimenting far longer. The limits are still real and instructive: 256 addresses means both the assembly and the C you can write stay small, and it's easy to run out of room — which is itself part of the lesson (see the note on memory limits below).

7.3 The machine

- **256 bytes of memory**, holding both program and data.
- **One accumulator** for arithmetic.
- **A program counter** that starts at 0.
- **14 opcodes**: load/store, add/sub, the bitwise operations, shifts, an unconditional `goto`, a single conditional `ifzero`, `nop`, and `stop`.

The original went through three versions: an experimental FreeDOS prototype, a Linux/ncurses prototype, and finally a FreeDOS graphics-mode program.

7.4 Building the original (and why v2 on macOS)

The three versions build very differently, which matters if you want to run Jim Hall's original alongside `toyasm`:

- **v3** (current main, the FreeDOS graphics-mode program) builds with a `.bat` script that calls **OpenWatcom C** (`wcl -q -2 -os toy.c ...`). That toolchain targets DOS; it does not build on macOS out of the box.
- **v2** (the Linux/ncurses prototype) ships a plain **Makefile** using `gcc` and `-lncurses`. That is the version to check out on macOS — it builds with `make + gcc` and a terminal ncurses UI, no OpenWatcom or DOS needed. The screenshot above is this v2 build.

So on macOS: check out v2 to actually build and run the original. v3 is FreeDOS/OpenWatcom-only.

7.5 What comes from `toycpu`, and what is original

Taken from `toycpu` (the instruction set):

- The opcode values are identical: `STOP=0`, `RIGHT=1`, `LEFT=2`, `NOT=15`, `AND=17`, `OR=18`, `XOR=19`, `LOAD=20`, `STORE=21`, `ADD=22`, `SUB=23`, `GOTO=24`, `IFZERO=25`, `NOP=128`.
- The `FETCH` bit (bit 4, `0x10`) marking instructions that carry an address byte.
- The execution semantics of each instruction, including the 8-bit wraparound on `ADD/SUB` (`>255 #sym.arrow #sym.minus 256, <0 #sym.arrow #sym.plus 256`).

This is unavoidable: matching the instruction set *means* matching these numbers and their behaviour. It is the specification, not copied code.

Written from scratch for `toyasm` (not present in `toycpu`):

- The two-pass **assembler** — labels, symbols, comments, data bytes. `toycpu` has no assembler; it reads pre-built binary via the switch-panel input shown above.
- The **Qt/PySide6 GUI debugger**. `toycpu` uses a DOS text-mode / ncurses display — a different language and toolkit entirely.

- The **CLI** verbose/step mode, the **EXPLAIN** output, the listing/export, and the `.toys/.toyo` file formats.

Both `toyasm`'s Python is deliberately **simple, single-file, standard-library + PySide6 code** — readability is a feature, since the point is to *see* how the machine works.

7.6 Licensing

Both projects are MIT-licensed, so the provenance is clean.

- `toycpu` is MIT, Copyright (c) 2022 Jim Hall (FreeDOS Project).
- `toyasm` is MIT, Copyright (c) 2026 Manfred Scheucher, for its own code (assembler, GUI, CLI).

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